

Ex. No: 09	Integration of Node.js with database module
12.03.2026	

Theme: Online event management system

Aim:

To integrate a Node.js backend with a MySQL database for storing and retrieving event registration data in an Online Event Management System..

Description:

The Online Event Management System integrates a Node.js backend with a MySQL database to enable efficient data handling and dynamic user interaction. The Node.js server, built using Express, acts as an intermediary between the frontend interface and the database. User registration data entered through the web interface is sent to the server using HTTP requests and stored in the database using SQL queries. The system also retrieves recent registration records from the database and displays them on the user interface in real time. Parameterized queries are used to ensure secure data insertion and prevent SQL injection. This integration allows seamless communication between different layers of the application, ensuring smooth data flow. Overall, the system demonstrates a functional full-stack architecture with effective backend-database connectivity.

Code:

Server.js

```
const mysql = require("mysql2");

const db = mysql.createConnection({
  host: process.env.DB_HOST,
  user: process.env.DB_USER,
  password: process.env.DB_PASSWORD,
  database: process.env.DB_NAME
});
```

INSERT Operation}});

```
app.post("/register", (req, res) => {  
  const { name, email, event } = req.body;  
  
  const sql = "INSERT INTO registrations (name, email, event) VALUES (?, ?, ?)";  
  
  db.query(sql, [name, email, event], (err) => {
```

SELECT Operation:

```
app.get("/recent", (req, res) => {  
  const sql = `  
    SELECT name, event, created_at  
    FROM registrations  
    ORDER BY created_at DESC  
    LIMIT 6  
  `;  
  
  db.query(sql, (err, results) => {
```

main.js:

```
fetch("http://localhost:3000/register", {  
  method: "POST",  
  headers: {  
    "Content-Type": "application/json"  
  },  
  body: JSON.stringify({  
    name,  
    email,  
    event: selectedEvent  
  })  
});
```

Fetching Data:

```
fetch("http://localhost:3000/recent");
```

Displaying Data in UI:

```
data.forEach(item => {  
  html += `  
    <h5>${item.name}</h5>  
    <p>${item.event}</p>  
  `;  
});
```

```
});
```

jquery-task.js:

```
$("#eventSearch").on("keyup", function () {
```

Input & Output:

```
E:\aiwd\backend>node server.js
◇ injected env (4) from .env //
Server running on port 3000 🚀
MySQL Connected ✅
```

View Registrations

santeep

★ Corporate Networking

🕒 6/4/2026, 7:08:00 pm

hariram

★ Technical Summit

🕒 6/4/2026, 7:01:50 pm

hjcj

★ Technical Summit

🕒 6/4/2026, 6:59:30 pm

Raj

suriya

Elgin

hospitalmanagment_db*

Limit to 1000 rows

```
7      event VARCHAR(100),
8      created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
9  );
10 • SHOW TABLES;
11
12 • SELECT * FROM registrations WHERE name = 'santeep';
13
14
```

Result Grid

	id	name	email	event	created_at
▶	6	santeep	santeep@karunya.edu.in	Corporate Networking	2026-04-06 19:08:00
*	NULL	NULL	NULL	NULL	NULL

Result Grid

Result:

The developed web-based event management system successfully enables users to register for events through an interactive interface. The application efficiently stores and retrieves registration data using a MySQL database. It provides accurate display of recent registrations, ensuring real-time data handling. Overall, the system meets the intended objectives of integrating frontend, backend, and database operations effectively.